

Experiment (11)

BALLISTIC PENDULUM

11.1 Introduction

We began this course with a study of kinematics, and defined simple quantities such as the velocity. How would one go about determining the velocity of a projectile? It is not straightforward to think of a direct method. Nonetheless, we can design a good demonstration using the conservation of momentum and conservation of energy. The ballistic pendulum is an example of an inelastic collision. Momentum is conserved during the collision, but energy conservation cannot be applied. After the collision, however, conservation of energy can be used in the swing of the combined masses upward, because gravity is the only external force on the system, and the gravitational potential energy is conservative. Using these facts, we will be able to “work backwards” to determine the original velocity of the projectile.

11.2 Objectives

- Investigate the conservation of mechanical energy. (Kinetic energy, potential energy, rotational kinetic energy).
- Investigate conservation of angular momentum.
- Use the ballistic pendulum to measure the initial projectile velocity.

11.3 Theory

The ballistic pendulum is a practical method to determine the velocity of a projectile. It consists of three parts as shown in Figure 11.1:

- A Ballistic unit (catapult).
- A ball that can be fired from the gun.
- A pendulum.
- A ball entry aperture at the end of the pendulum to catch the ball.

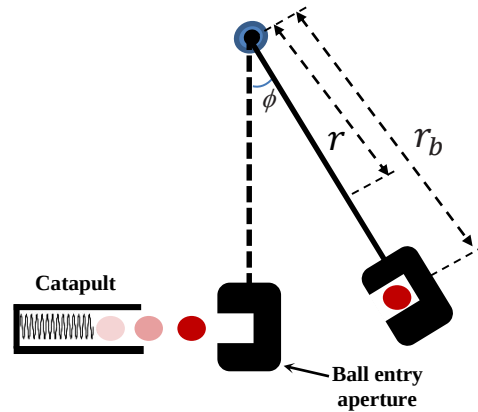


Figure 11.1: Experimental Setup diagram.

When the catapult fires a ball of mass m_b as a projectile, it acquires an initial velocity v_i . The ball entry aperture catches the ball and the pendulum swings up by a deflection angle ϕ .

To measure the velocity of the fired ball we apply two physical identities. The first is derived from the conservation of momentum before and after catching the ball. The second is the conservation of energy of the swinging pendulum in terms of the swing time and the deflection angle ϕ .



Note that the collision between the ball and pendulum is inelastic so that the kinetic energy is not conserved. We cannot equate the kinetic energy of the combined system of the pendulum and the ball just after collision with the kinetic energy of the ball just before the collision.

There are two ways for calculating the initial velocity of the ball (See the [Appendix](#) the derivation).

11.3.1 The approximate method

In the approximate method, the combined pendulum and ball are considered as a point mass located at their center of mass. This method is an approximation as it does not consider the rotational inertia of the pendulum but is somewhat easier. Using the approximation, the velocity of the projectile is given by

$$v_{app} = \frac{m + M}{m} \sqrt{2 g r (1 - \cos \phi)} \quad (11.1)$$

where m and M are the masses of the projectile and the pendulum respectively, g is the gravitational acceleration, r the distance between the pendulum pivot and the center of gravity of the combined pendulum and the captured ball (this radius is *not* in general equal to r_b , which is the distance from the axis of rotation of the pendulum to the center of the ball).

11.3.2 The exact method

This method incorporates the rotational kinetic energy of the pendulum. Here, the velocity of the ball is given by

$$v_{ex} = \frac{m + M}{m} \frac{r}{r_b} \frac{g T}{2\pi} \sqrt{2 (1 - \cos \phi)} \quad (11.2)$$

where m and M are the masses of the projectile ball and the pendulum respectively, g is the gravitational acceleration, r is the distance between the axis of rotation and the center of gravity of the combined pendulum and captured ball, r_b is the distance between the axis of rotation and the center of the captured ball, and T is the oscillation period of the combined pendulum and captured ball and is related to the length of the equivalent mathematical pendulum r_m by the relation

$$T = 2\pi \sqrt{\frac{r_m}{g}} \quad (11.3)$$

11.4 Setup

The Ballistic Pendulum main parts are shown in Figure 11.2.



Figure 11.2: Ballistic Pendulum main parts.

- | | |
|--------------------------------------|--------------------------|
| 1 Ballistic unit (catapult) | 5 Speed measuring device |
| 1a Release lever | 6 Ball storing tube |
| 1b Tension lever | 7 Thread |
| 2 Projectiles (wooden balls) | 8 Stop watch |
| 3 Angle scale | 9 Digital balance |
| 4 Pendulum with the trailing pointer | 10 Measuring tape |

11.5 Procedure

11.5.1 Task (1): Determination of the initial velocity of the projectile ball using the speed measuring device, the approximate and the exact methods.

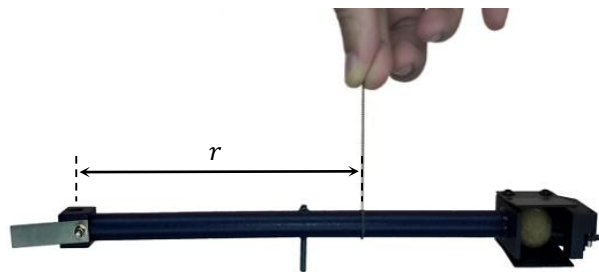
- 1- Determine the mass M of the pendulum (4), by removing the knurled screw as shown in Figure 11.3. Take off the pendulum rod and weigh it using the digital balance (9).
- 2- Weigh the wooden ball (2) using the digital balance.
- 3- Place the wooden ball inside the ball entry aperture of the pendulum as shown in Figure 11.4 (a). Subsequently, determine the center of gravity of the combined system by suspending the pendulum rod with the ball by using a loop of thread (7) as shown in Figure 11.4 (b).
- 4- Move the loop hole back and forth along the pendulum rod until the equilibrium position is achieved, i.e, the system is horizontally balanced under the force of gravity as shown in Figure 11.4 (b).



Figure 11.3: Removing of the knurled screw.



(a)



(b)

Figure 11.4: Placing the ball inside the ball entry aperture of the pendulum and determining of the center of gravity of the combined system.

- 5- Measure the distance r between the suspension point (center of gravity) and the axis of the pendulum using the measuring tape (10) as shown in Figure 11.4 above.
- 6- Fix the pendulum rod back into its original place using the knurled screw.
- 7- To measure the pendulum's oscillation period, T , the pendulum spring can be temporarily turned toward the front until the movement is no longer blocked by the pin as shown in Figure 11.5 (a). Turn the trailing pointer away from the pendulum as shown Figure 11.5 (b).

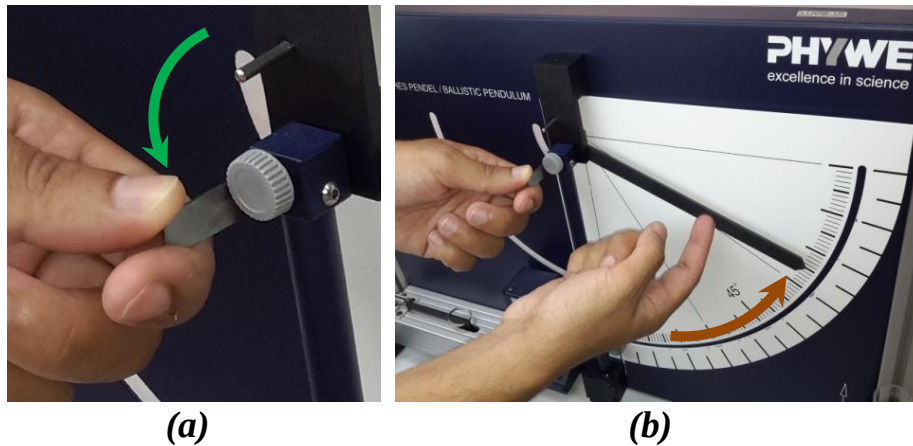


Figure 11.5: Tuning the pendulum spring toward the front and the trailing pointer away from the pendulum.

- 8- Place the wooden ball inside the ball entry aperture of the pendulum.
- 9- Push the end of the pendulum towards the speed measuring device as shown in Figure 11.6. Let the pendulum go, it will start oscillating. Using the stopwatch (8), measure the time of five periods of the pendulum's oscillations T_5 (make sure the pendulum does not collide with the speed measuring device and is free to oscillate).



Figure 11.6: Measuring the pendulum's oscillation period.

- 10- Return the pendulum spring and the trailing pointer back to their original position. (Reverse what has been done in step 7)



Failing to **return the pendulum spring** to its original position will cause the pendulum rod to **collide the speed measuring device after shooting the ball**. With time, such collisions may cause the pendulum to break.

- 11- While the catapult (1) is released, take the wooden ball and attach it to the catapult as shown in Figure 11.7 below.



Put the wooden ball in the correct position as shown in Figure 11.7.



Figure 11.7: Correct and incorrect ways to attach the ball to the catapult.

- 12- Pull the catapult tension lever (1a) back to the first position (there are three positions) until the lock-in position is reached and you hear a click.
- 13- Reset the speed measuring device by pressing its “Reset” button as shown in Figure 11.8



Figure 11.8: Resetting the speed measuring device.

- 14- Ensure that the pendulum is at rest and the trailing pointer (4) indicates zero on the angle scale.
- 15- Trigger the shot by pulling the release lever (1b).



On pulling the release lever, do not place your hand near the opening of the catapult nor in the direction of motion of the fired ball. Otherwise, serious injury may occur.

- 16- Record the ball speed measured by the speed measuring device.



The maximum deflection angle ϕ cannot be measured after shooting the ball once. Why? Due to friction involved in the motion of the trailing pointer, the measured deflection angle can be less than the correct value. It is therefore important to shoot the ball at the resting pendulum using the same catapult tension for a second and a third time without resetting the trailing pointer. It will be noticed that the deflection of the trailing pointer increases for almost every shot.

- 17- Repeat steps 15 and 16 another four times.
- 18- After the fourth repetition of step 17 (shooting the ball five times) the deflection angle ϕ can be considered maximum and will not change if the ball is shot again (one can assume that this final deflection angle is not falsified by friction). Read this maximum deflection angle using the trailing pointer and the angle scale (3).
- 19- Repeat steps 11 to 18 another two times, once when the catapult tension lever (in step 12) is at the second position and another when the catapult tension lever is at the third position.

- 20- Using relation (11.1) and relation (11.2). Calculate the approximated initial velocity v_{app} and the exact initial velocity v_{ex} , respectively, of the projectile for every maximum deflection angle obtained from setting the catapult tension lever to the three different positions.
- 21- Compare your result to the speed of the projectile measured using the speed measuring device.
- 22- Plot the projectile velocity as a function of the maximum deflection angle using the approximate formula (11.1), the exact formula (11.2), and by using the direct measurement of a speed measuring device, all on the same graph.
- 23- Make a comparison between the three curves.

Appendix

1- Approximate Method

Let v_b be the speed of the ball just before collision, and v_p the speed of the pendulum and ball just after collision, then, the initial kinetic energy of the ball and pendulum just after collision is given by,

$$KE = \frac{1}{2} (M + m)v_p^2 \quad (11.4)$$

the potential energy of the pendulum at the top of its swing,

$$PE = (M+m) g \Delta h_c \quad (11.5)$$

where $\Delta h_c = r(1 - \cos \theta)$, is the change in height, so,

$$\Delta PE = (M+m) g r (1 - \cos \phi) \quad (11.6)$$

This potential energy gained is equal to the kinetic energy lost between the moment of collision and the time that the pendulum reaches the top of its swing, i.e.

$$\Delta PE = \Delta KE \quad (11.7)$$

$$\Delta PE = (M + m)gr (1 - \cos \phi) = \frac{1}{2} (M + m)v_p^2 \quad (11.8)$$

from which we get,

$$v_p = \sqrt{2gr (1 - \cos \phi)} \quad (11.9)$$

but from conservation of linear momentum before and after collision and neglecting the rotational inertia, we have,

$$mv_b = (M+m) v_p \quad (11.10)$$

where v_p is the velocity of the combination just after the collision. From equations (11.9) and (11.10) we get,

$$v_b = \frac{(M + m)}{m} \sqrt{2gr (1 - \cos \phi)} \quad (11.11)$$

Note that the method is approximate because we ignored the fact that the rod connecting the pendulum to the pivot has rotational kinetic energy immediately after the collision. Under what conditions will that be a good approximation?

2- Exact Method

In this case, the potential energy is found as in equation (11.5). For the equation of kinetic energy, we use the equation for rotational kinetic energy instead of the linear kinetic energy.

Let I be the moment of inertia of the combination of the ball and pendulum, and ω be the angular velocity immediately after the collision then, the rotational kinetic energy is given by,

$$KE_R = \frac{1}{2} I \omega^2 \quad (11.12)$$

and this is equal to the potential energy, i.e

$$\frac{1}{2} I \omega^2 = (M + m) gr (1 - \cos \phi) \quad (11.13)$$

The angular momentum of the ball before the collision as measured from the pendulum pivot point $= I_b \omega_b = (mr_b^2) \left(\frac{v_b}{r_b} \right) = m r_b v_b$

And the angular momentum of the combination just after the collision as measured from the pendulum pivot point $= I \omega$

From conservation of angular momentum before and after collision, and we have,

$$m r_b v_b = I \omega \quad (11.14)$$

$$\omega = \frac{m r_b v_b}{I}$$

From equations (11.13) and (11.14) we get,

$$v_b = \frac{1}{m r_b} \sqrt{2 I g r (M + m)(1 - \cos \phi)} \quad (11.15)$$

The value of I can be obtained in terms of the periodic time of the pendulum. To do this, we start with the rotational equivalent of Newton's second law,

$$\tau = I \alpha \quad (11.16)$$

where τ is torque, I is moment of inertia, and α is angular acceleration.

The component of that force directed towards the center of the pendulum swing is given by,

$$F = - (M + m) g \sin \phi \quad (11.17)$$

and the torque on the pendulum is thus,

$$I \alpha = - r (M + m) g \sin \phi \quad (11.18)$$

For small angles ϕ , $\sin \phi \approx \phi$, so if we make this substitution and solve for α we get:

$$\alpha = \frac{d^2 \phi}{dt^2} = - \frac{(M + m) g r}{I} \phi \quad (11.19)$$

This angular equation is in the same form as the equation for linear simple harmonic motion:

$$\frac{d^2 \phi}{dt^2} = - \omega^2 \phi \quad (11.20)$$

Comparing these two equations, we get,

$$\omega^2 = \frac{(M + m)gr}{I} \quad (11.21)$$

$$I = \frac{(M + m)gr}{\omega^2}$$

But $\omega = \frac{2\pi}{T}$, where T is the periodic time of the pendulum, so,

$$I = (M + m)gr\left(\frac{T}{2\pi}\right)^2 \quad (11.22)$$

And substituting in equation (11.15) we get,

$$v_b = \frac{m + M}{m} \frac{r}{r_b} \frac{g T}{2\pi} \sqrt{2(1 - \cos\phi)} \quad (11.23)$$

Note that we have made a small-angle approximation to find this equation for I ; but I does not depend on ϕ . This means that we must first measure the period T using small oscillations, then, we may use that value of I regardless of the amplitude reached during other parts of the experiment.